I. Execution and Native Feedback

A. From physical release to supported service

We isolate computation retention R and coverage exit K while enabling the legal two-lane Consumer S in every arm. The study uses the same executor, 16 fixed development sources, and 250.180 s of offered audio per run. Four arms, synchronized or staggered arrivals, and three fresh-process repetitions give 24 runs. A frozen wrong-conditioning fixture supplies controlled useless tails. It is separate from the unmodified-input work used to establish lawful continuation. Original input obligations, D2/C8 deadlines, quality, and physical costs remain unchanged.

All selected Canonical $F/Z/P/T$ histories match, with WER $14/687$ in every run. Across the paired runs, all 675 pairs with identical complete processed inputs, actual prompt/audio identities, and peer order return identical complete outputs. This matching distinguishes an earlier execution opportunity from a comparison between different batches.

Each synchronized $K1$ run releases a following batch of 16 rows that matches its $K0$ counterpart on these inputs and outputs. Under $R1$, the following model computation starts 473/577/498 ms earlier across the three repetitions. Request, acknowledgment, and drain remain distinct: in one repeat, request-to-ACK is 22.196 ms and ACK-to-actual-drain is 13.359 ms. All 96 following rows across the six synchronized comparisons are actually consumed and first advance their target frontiers.

An obsolete proposal in $K0$ does not itself establish a missed input obligation: Canonical may already have covered it. Comparing the first actual view from either lane at each target, only 6/6/1 rows under $R1$ and 4/2/6 under $R0$ change from dual-deadline miss to hit. The other rows were already timely. A separate full-text audit confirms that all 96 first-consumed hypotheses agree byte-for-byte with the complete Canonical text selected later at exactly the same frontier. Their actual worker records have no input override. Thus the newly timely rows retain complete content support; this is retrospective selected-model agreement, not an online content certificate or transcription groundtruth.

The whole-run boundary is less uniform. Adding K to $R1$ increases original credit by 15.255/13.265/7.970 audio seconds under synchronized arrivals, but changes it by $-26.000/-6.475/+21.780$ s under staggered arrivals. Neither whole-run model time nor board energy improves consistently. Whole-run first content also remains revisable: each synchronized arm has 16 s of timely first-ahead content differing from later same-frontier Canonical text, or 14 s after the fixed normalizer. These equal deductions leave the synchronized credit differences intact. We therefore use simple coverage exit as execution support, with a measured drain-to-service chain, rather than a universally beneficial cancellation algorithm or a stable positive $R \times K$ interaction.

B. Native feedback correction and its capacity boundary

We additionally connect the published StreamCorrect recognizer and learned corrector [1]. Its chosen correction feeds subsequent recognition. We preserve this original selection and feedback update, use the released checkpoints, and declare the fixed 500 ms/no-VAD front end as a controlled adaptation. Four metadata-selected utterances form two two-source cohorts. Two natural-interleaving witnesses precede the frozen formal comparison: three arms, two cohorts, and two fresh processes per arm/cohort, giving 12 formal runs.

TABLE I

Native formal runs. Each arm covers the same 66.42975 s of repeated offered audio. Credit uses the original D2/C8 deadlines; C99 is the range across four runs. Energy is the sum over both boards from the original service origin through actual Consumer drain, excluding model loading and warmup. No arm meets both whole-run P99 targets in any run.

Arm	Credit (s)	Energy (kJ)	C99 (s)
Two replicas	18.5	25.949	5.08–8.38
Discard branches	12.5	49.784	16.67–19.15
Retain branches	17.0	47.991	16.95–21.41

The native adapter compares complete closed feedback cuts: audio buffers, token groups, context, alignment position, language/configuration, and frontier. The declared boundary requires cleaned native decoding caches. Initial buffer-only updates can advance this state without earning processed-audio service credit. A local FIFO Consumer records actual applications. This adaptation exercises the full-state dependency and consumption relation using its own reconciler and journals; it does not reuse the core event kernel or fixed-batch retirement wrapper, whose recovery and drain properties are evaluated separately.

The sequential arm is a strong resource control: each GPU hosts a complete recognizer–corrector replica for one source. Discard and retain instead use one Authority executing the complete original pipeline and one Private recognizer predicting the first-beam continuation through the native feedback-update tail. They share the Consumer. The discard arm clears old branches conservatively; the retain arm preserves compatible branches. Private results never replace the original corrector's selection.

Within each cohort, complete Canonical state/output histories match across the three arms and both repetitions. All arms have 16 character edits over 182 reference characters, giving 8.79% CER under the adaptation's whitespace-removal rule. This is the quality of the fixed adaptation, not a reproduction of the author's benchmark CER. The retain arm encounters 87 compatible completed-ahead boundaries and 13 incompatible ones. It discards the incompatible branches; separately counted actual visible-frontier retractions total 13. These decisions use complete state rather than surface text change. Compatible retention does not imply that every provisional view should persist.

Table I exposes a real cost boundary. Retention adds 4.5 s of credit over discard and reduces its board energy by 3.60%, but has less credit than two complete replicas and consumes 84.94% more board energy. Cohort effects differ: retain-minus-

replicas credit is $+3.5/+1.5\,\mathrm{s}$ in A and $-4.0/-2.5\,\mathrm{s}$ in B. The learned-feedback application confirms actual compatible and incompatible continuations; its tested two-stream workload also shows that preserving Private work does not remove the capacity cost of an independently advancing Authority.

REFERENCES

[1] T. D. Vu, Z. Xie, W. Zhao, V. J. Wei, D. Jiang, and Y. R. Fung, "StreamCorrect: Self-supervised error correction for real-time ASR with offline recognizers," in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing: Industry Track*. Association for Computational Linguistics, Oct. 2026, pp. 2391–2403. [Online]. Available: https://aclanthology.org/2026.emnlp-industry.142/